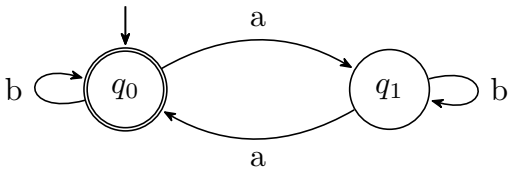
オートマトン・言語理論 レポート課題 01

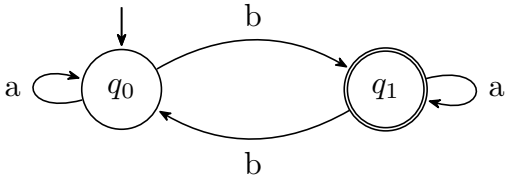
C5TB2006 安倍大陽

2026 年 10 月 18 日

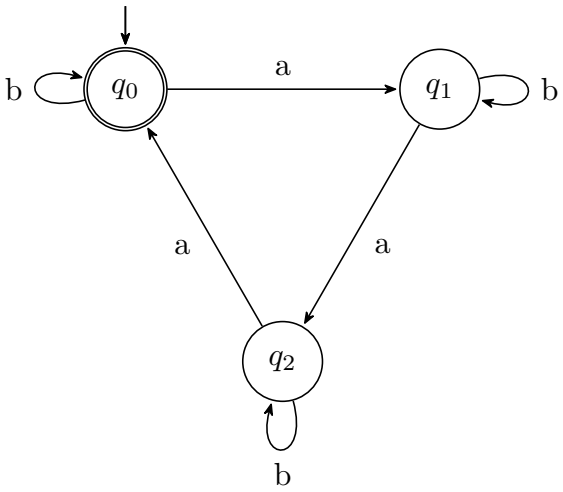
0.1 $L_1 = \{w \in \Sigma^* \mid \langle w \rangle_a = 2k, k \in \mathbb{N}\}$



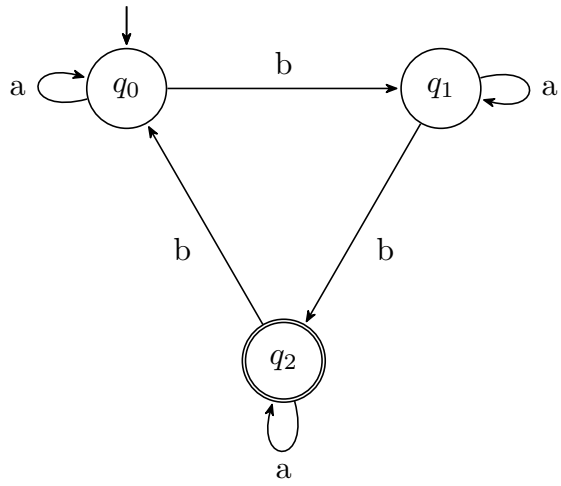
0.2 $L_2 = \{w \in \Sigma^* \mid \langle w \rangle_b = 2k + 1, k \in \mathbb{N}\}$



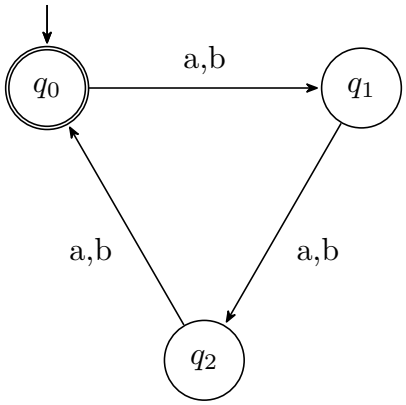
0.3 $L_3 = \{w \in \Sigma^* \mid \langle w \rangle_a = 3k, k \in \mathbb{N}\}$



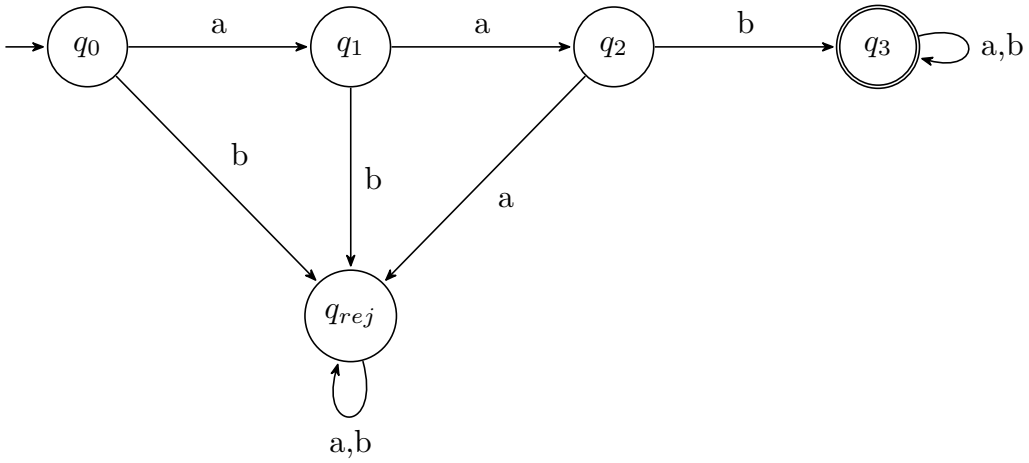
0.4 $L_4 = \{w \in \Sigma^* \mid \langle w \rangle_b = 3k + 2, k \in \mathbb{N}\}$



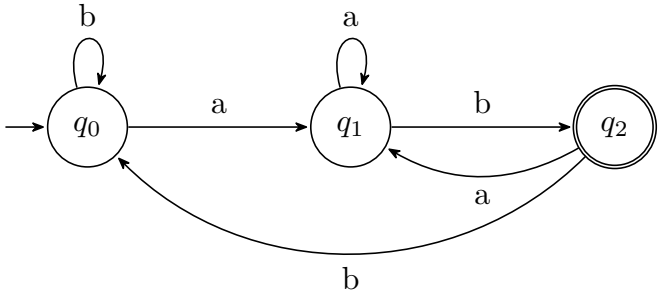
0.5 $L_5 = \{w \in \Sigma^* \mid |w| = 3k, k \in \mathbb{N}\}$



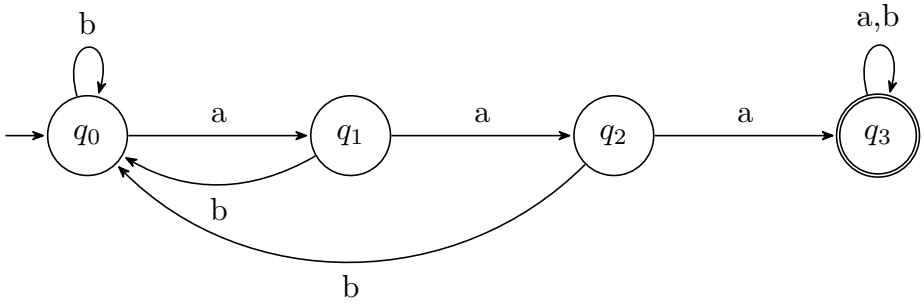
0.6 $L_6 = \{\mathsf{aab}v \mid v \in \Sigma^*\}$



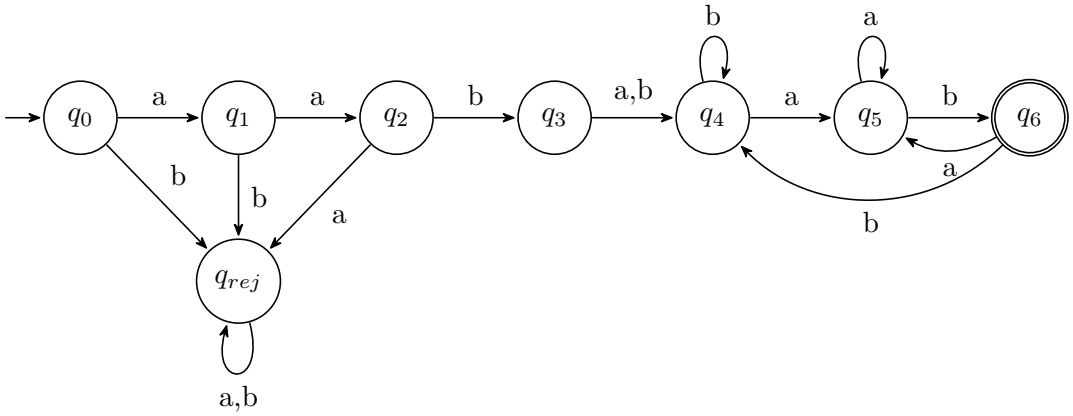
0.7 $L_7 = \{u\mathsf{ab} \mid u \in \Sigma^*\}$



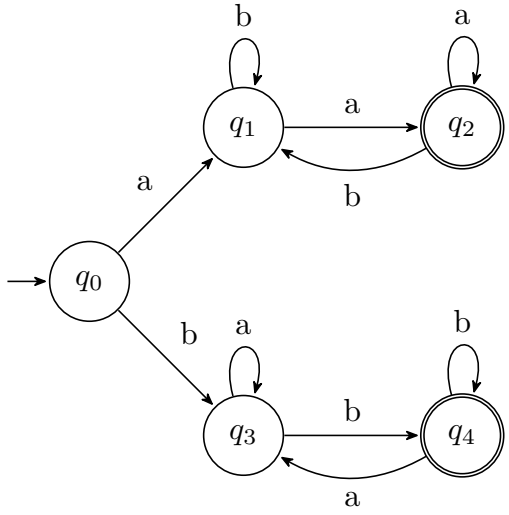
0.8 $L_8 = \{u\mathsf{aaa}v \mid u, v \in \Sigma^*\}$



0.9 $L_9 = \{\mathsf{aab}v\mathsf{ab} \mid v \in \Sigma^*\}$



0.10 $L_{10} = \{\mathsf{a}v\mathsf{a} \mid v \in \Sigma^*\} \cup \{\mathsf{b}v\mathsf{b} \mid v \in \Sigma^*\}$



0.11 $L_{11} = \{\mathsf{a}^i\mathsf{b}^j \mid i \geq 0, j \geq 0\}$